

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Solve the quadratic equation $x^2 = 81$.

☐ A $x = 9$ only

☐ B $x = -9$ only

☐ C $x = \pm 81$

☐ D $x = \pm 9$

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Q2 · Rank 1 / Foundation

Solve $f(x) = 0$ for $f(x) = x(100 - x)$.

☐ A $x = 0$ or $x = 100$

☐ B $x = 100$ only

☐ C $x = 0$ only

☐ D $x = -100$ or $x = 100$

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Q3 · Rank 1 / Foundation

If the demand function of a good is given by $P = 80 - 3Q$, find the price when $Q = 10$, and deduce the total revenue.

☐ A $P = 50, TR = 50$

☐ B $P = 50, TR = 500$

☐ C $P = 77, TR = 770$

☐ D $P = 110, TR = 1100$

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Q4 · Rank 2 / Core

Given the demand function $P = 10 - 4Q$, express TR as a function of Q .

☐ A $TR = 10 - 4Q$

☐ B $TR = 10Q - 4Q$

☐ C $TR = 10Q - 4Q^2$

☐ D $TR = 10Q + 4Q^2$

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Q5 · Rank 2 / Core

If fixed costs are 100 and variable costs are 5 per unit, find the total cost when $Q = 10$.

☐ A 50☐ B 105☐ C 1000☐ D 150

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Q6 · Rank 2 / Core

For demand $P = 80 - 3Q$, fixed costs 100 and variable costs 5 per unit, use the revenue and cost when $Q = 10$ to work out the corresponding profit.

☐ A 350☐ B 500☐ C 150☐ D 650

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Q7 · Rank 3 / Application

Given the total revenue function $TR = 50Q - 4Q^2$, find the corresponding demand function. Assume $Q > 0$.

☐ A $P = 50Q - 4Q^2$

☐ B $P = 50 - 4Q$

☐ C $P = 50 - 8Q$

☐ D $P = 50 + 4Q$

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Q8 · Rank 3 / Application

The total cost, TC , of producing 100 units of a good is 600 and the total cost of producing 150 units is 850. Assuming that the total cost function is linear, find an expression for TC in terms of Q , the number of units produced.

☐ A $TC = 6Q$

☐ B $TC = 5Q$

☐ C $TC = 100 + 5Q$

☐ D $TC = 500 + Q$

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Q9 · Rank 3 / Application

Given that fixed costs are 500 and variable costs are 10 per unit, express AC as a function of Q . Assume $Q > 0$.

☐ A $AC = 500 + 10Q$

☐ B $AC = 500/Q$

☐ C $AC = 10Q$

☐ D $AC = 500/Q + 10$

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Q10 · Rank 4 / Challenge · OPTIONAL

Given the quadratic supply and demand functions $P = Q_S^2 + 2Q_S + 12$ and $P = -Q_D^2 - 4Q_D + 68$, determine the equilibrium price and quantity. Use nonnegative quantity.

☐ A $P = 36, Q = 4$

☐ B $P = 27, Q = -7$

☐ C $P = 4, Q = 36$

☐ D $P = 68, Q = 0$

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